

Advanced (Habanero)

- Q1.** What is the purpose of graphing multiple inequalities on the same coordinate plane?
- (A) To find the intersection point
 (B) To identify overlapping solution regions
 (C) To determine the slope of a line
 (D) To find the y -intercept
 (E) To solve a system of linear equations
- Q2.** Which of the following is a feasible region for the system of inequalities $x + y \geq 2$ and $x - y \leq 1$?
- (A) The region above the line $x + y = 2$ and below the line $x - y = 1$
 (B) The region below the line $x + y = 2$ and above the line $x - y = 1$
 (C) The region above the line $x + y = 2$ and above the line $x - y = 1$
 (D) The region below the line $x + y = 2$ and below the line $x - y = 1$
 (E) The region on the line $x + y = 2$ and on the line $x - y = 1$
- Q3.** What is the result of graphing the inequalities $x \geq 2$ and $y \leq 3$ on the same coordinate plane?
- (A) A line with a slope of 1
 (B) A line with a slope of -1
 (C) A vertical line at $x = 2$ and a horizontal line at $y = 3$
 (D) A circle with center $(2, 3)$
 (E) A parabola with vertex $(2, 3)$
- Q4.** How many solution regions are formed by the system of inequalities $x + y \geq 1$ and $x - y \leq 2$?
- (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) 5
- Q5.** Which of the following systems of inequalities has no solution?
- (A) $x + y \geq 2$ and $x + y \leq 1$
 (B) $x - y \geq 1$ and $x - y \leq 2$
 (C) $x \geq 2$ and $x \leq 1$
 (D) $y \geq 3$ and $y \leq 2$
 (E) $x + y \geq 1$ and $x - y \leq 2$
- Q6.** What is the equation of the boundary line for the inequality $x + 2y \geq 3$?
- (A) $x + 2y = 3$
 (B) $x - 2y = 3$
 (C) $2x + y = 3$
 (D) $2x - y = 3$
 (E) $x - y = 3$
- Q7.** Which of the following points lies within the feasible region of the system of inequalities $x + y \geq 2$ and $x - y \leq 1$?
- (A) $(1, 1)$
 (B) $(2, 2)$
 (C) $(3, 1)$
 (D) $(1, 3)$
 (E) $(2, 0)$
- Q8.** How many corner points does the feasible region of the system of inequalities $x + y \geq 1$ and $x - y \leq 2$ have?
- (A) 2
 (B) 3
 (C) 4
 (D) 5
 (E) 6

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y \geq 2$ and $x - y \leq 1$. Identify the corner points of the feasible region and determine the equation of the boundary lines.
- Q10.** Solve the system of inequalities $x + 2y \geq 3$ and $x - y \leq 2$. Determine the feasible region and identify any corner points. Provide a sketch of the graph.

Chef's Tips

- Ensure students use a pencil and graph paper for graphing.
- Encourage students to label the axes and boundary lines clearly.
- Remind students to check their work by plugging in test points.